
Report 2

Simulation of diffusion processes

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Framework

We now consider the new CEV model given by :

$$(\mathbf{CEV}) : \begin{cases} dX_t = \sigma_0 X_t^\beta dW_t \\ X_0 = x \end{cases}$$

where $\sigma_0 > 0$ and $\beta \in]0, 1]$. If $\beta = 1$, the model corresponds to Black-Scholes :

$$(\mathbf{BS}) : \begin{cases} dX_t^{\mathbf{BS}} = \sigma_0 X_t^{\mathbf{BS}} dW_t \\ X_0^{\mathbf{BS}} = x \end{cases}$$

The main purpose of this second report is to build a new estimator for $\mathbb{E}^{\mathbb{P}} \left[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}} \right]$ where X follows **(CEV)**, by using the results of the first report on **(BS)**.

More accurately, if we define the previous barrier european call price under Black-Scholes by :

$$C^{\mathbf{BS}}(T, x) = \mathbb{E}^{\mathbb{P}} \left[(X_T^{\mathbf{BS}} - K)_+ \cdot \mathbb{1}_{\{\min_{0 \leq t \leq T} X_t^{\mathbf{BS}} > L\}} \right]$$

we need to show the following equality :

$$\begin{aligned} & \mathbb{E}^{\mathbb{P}} \left[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}} \right] \\ &= C^{\mathbf{BS}}(T, x) + \int_0^T \mathbb{E}^{\mathbb{P}} \left[(\mathcal{L} - \mathcal{L}^{\mathbf{BS}}) C^{\mathbf{BS}}(T - s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}} \right] ds \end{aligned}$$

where $\mathcal{L}$ and $\mathcal{L}^{\mathbf{BS}}$ are called the generators of **(CEV)** and **(BS)** respectively.

1. $\mathcal{L}$ and $\mathcal{L}^{\text{BS}}$ generators computation

Let's start by computing the two generators. For the record :

Theorem : Generator

A generator is an operator $\mathcal{L} : f \rightarrow \mathcal{L}f$ defined as follows :

$$\frac{\mathbb{E}^{\mathbb{P}} \left[f \left(X_t^{(\mathbf{M})} \right) \right] - f(x)}{t} \xrightarrow[t \rightarrow 0]{} \mathcal{L}f$$

where $(X_t)_t$ is the solution of a given stochastic model $(\mathbf{M})$ of starting point x . $\mathcal{L}$ is called the generator of $(\mathbf{M})$.

So, for (CEV) , we have by Itô :

$$\begin{aligned} df(X_t) &= f'(x_t)dX_t + \frac{1}{2}f''(X_t)\sigma_0^2(X_t)dt \\ &= f'(X_t)\sigma_0 X_t^\beta dW_t + \frac{1}{2}\sigma_0^2 X_t^{2\beta} f''(X_t)dt \\ \Rightarrow f(X_t) &= f(x) + \int_0^t f'(X_s)\sigma_0 X_s^\beta dW_s + \frac{1}{2} \int_0^t \sigma_0^2 X_s^{2\beta} f''(X_s)ds \end{aligned}$$

With a reasonable f , because $\beta \leq 1$,

$$\mathbb{E}^{\mathbb{P}} \left[\int_0^t (f'(X_s)\sigma_0 X_s^\beta)^2 ds \right] < +\infty, \quad \forall t$$

so $\left(\int_0^t f'(X_s)\sigma_0 X_s^\beta dW_s \right)_{t \in [0, T]}$ is a martingale. Consequently, we have :

$$\mathbb{E}^{\mathbb{P}} [f(X_t)] = f(x) + \frac{1}{2} \int_0^t \sigma_0^2 \mathbb{E}^{\mathbb{P}} [X_s^{2\beta} f''(X_s)] ds$$

Note : to exchange the $\int$ and $\mathbb{E}$ signs, we need (Fubini) f to verify $\mathbb{E}^{\mathbb{P}} \left[\int_0^t |X_s^{2\beta} f''(X_s)| ds \right] < +\infty \quad \forall t$.

Under those conditions, we can now compute $\mathcal{L}f$ and $\mathcal{L}^{\text{BS}}f$ as :

$$\frac{\mathbb{E}^{\mathbb{P}} [f(X_t)] - f(x)}{t} = \frac{1}{t} \int_0^t \frac{1}{2} \sigma_0^2 \mathbb{E}^{\mathbb{P}} [X_s^{2\beta} f''(X_s)] ds \xrightarrow[t \rightarrow 0]{} \frac{1}{2} \sigma_0^2 x^{2\beta} f''(x)$$

So eventually :

$$\mathcal{L}f = \frac{1}{2}\sigma_0^2 x^{2\beta} f''(x)$$

and

$$\mathcal{L}^{\text{BS}} f = \frac{1}{2}\sigma_0^2 x^2 f''(x)$$

2. Clarifying C^{BS}

Thanks to the previous part, we know that we need C^{BS} to be twice differentiable, but is it the case ? The point of this part is to give a proper form to $C^{\text{BS}}(\tau, x)$.

Let's start :

$$\begin{aligned} C^{\text{BS}}(\tau, x) &= \mathbb{E}^{\mathbb{P}} \left[(X_T^{\text{BS}} - K)_+ \mathbb{1}_{\{\min_{0 \leq t \leq \tau} X_t^{\text{BS}} > L\}} \right] \\ &= \int_0^{+\infty} \int_0^{+\infty} (y - K)_+ \mathbb{1}_{\{l > L\}} p_{\tau}^x(y, l) \, dy dl \end{aligned}$$

(where $p_{\tau}^x(\cdot, \cdot)$ is the joint density of $(X_{\tau}, \min_{t \leq \tau} X_t)$ conditioned to $X_0 = x$)

$$\begin{aligned} &= \int_L^{+\infty} \int_K^{+\infty} (y - K) p_{\tau}^x(y, l) \, dy dl \\ &= \int_K^{+\infty} (y - K) \left(\int_L^{+\infty} p_{\tau}^x(y, l) dl \right) dy \\ &:= \int_K^{+\infty} (y - K) p_{\tau}^x(y) dy \end{aligned}$$

where we define $p_{\tau}^x(\cdot)$ as the density of $X_{\tau} | X_0 = x$ constrained to $\{\min_{0 \leq t \leq \tau} X_t > L\}$. p_{τ}^x being the only x dependant function in C^{BS} , differentiating C^{BS} with respect to x requires us to compute p_{τ}^x , which is not an easy task. On the other hand, we proved in chapter I that a change of variable and probability can give us a new process Y that is a Brownian motion. That is the possibility we explore in next part.

3. Using Y instead of X

A lot of result used in this part are proved in Chapter I, particularly for the Lamperti transform $Y = g(X)$ and for the change of probability $\mathbb{P} \longleftrightarrow \mathbb{Q}$ by Girsanov. These two transformations allows us to do the next derivation :

$$\begin{aligned} C^{\text{BS}}(\tau, x) &= \mathbb{E}^{\mathbb{P}} \left[(X_{\tau}^{\text{BS}} - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} X^{\text{BS}} > L\}} \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[(g^{-1}(Y_{\tau}) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \right] \\ &= \mathbb{E}^{\mathbb{Q}} \left[(g^{-1}(Y_{\tau}) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \cdot \frac{d\mathbb{P}}{d\mathbb{Q}} \right] \end{aligned}$$

$$\text{with } \frac{d\mathbb{P}}{d\mathbb{Q}} = \frac{1}{L_{\tau}} \text{ (see chapter I for } L_t \text{ definition)}$$

$$\begin{aligned} &= \exp \left(\int_0^{\tau} a(Y_s) dW_s + \frac{1}{2} \int_0^{\tau} a^2(Y_s) ds \right) \\ &= e^{-\frac{1}{2}\sigma_0 W_{\tau} + \frac{1}{8}\sigma_0^2 \tau} \text{ since } a(y) = -\frac{1}{2}\sigma_0 \\ &= e^{-\frac{1}{2}\sigma_0(Y_{\tau} - g(x)) - \frac{1}{8}\sigma_0^2 \tau} \text{ as } dY_t = -\frac{1}{2}\sigma_0 dt + dW_t \\ &:= w(Y_{\tau}, \tau) \end{aligned}$$

$$\begin{aligned} \text{So } C^{\text{BS}}(\tau, x) &= \mathbb{E}^{\mathbb{Q}} \left[(g^{-1}(Y_{\tau}) - K)_+ \mathbb{1}_{\{\min_{t \leq \tau} Y_t > g(L)\}} \cdot w(Y_{\tau}, \tau) \right] \\ &= \int_{g(K)}^{+\infty} \int_{g(L)}^{+\infty} (g^{-1}(u) - K) w(u, \tau) q_{\tau}^x(u, l) \, du dl \end{aligned}$$

where we define $q_{\tau}^x(\cdot, \cdot)$ as the joint density of $(Y_{\tau}, \min_{t \leq \tau} Y_t)$ conditioned to $Y_0 = g(x)$. This time, the density is much easier to compute. We know from Chapter I that :

$$q(t, s, a, b) = \mathbb{Q} \left(\min_{r \in [t, s]} Y_r \leq g(L) \mid Y_t = g(a), Y_s = g(b) \right)$$

which is the cumulative distribution function of $(\min_r Y \mid Y_t, Y_s)$

$$\implies \frac{dq}{d(g(L))}(t, s, a, b) = \tilde{q}(g(L))$$

with $\tilde{q}$ the density of $(\min_r Y \mid Y_t, Y_s)$

$$\Rightarrow f_s(g(b)) \cdot \frac{dq}{d(g(L))}(t, s, a, b) = \tilde{q}(g(L)) \cdot f_s(g(b))$$

If f_s is defined as the density of Y_s , $\tilde{q}(g(L)) \cdot f_s(g(b))$ is exactly the density of $(\min_r Y_r, Y_s)$ conditionally to $Y_t = g(a)$, which is $q_\tau^x(\cdot, \cdot)$. Consequently :

$$\begin{aligned} q_\tau^x(u, l) &= f_{\tau(u)} \cdot \frac{d}{dl} q(0, \tau, x, g^{-1}(u)) \text{ valid for } l \leq \min(g(x), u) \\ &= \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \frac{d}{dl} \left(e^{-\frac{2}{\tau}(l-g(x))(l-u)} \right) \\ &= -\frac{2}{\tau} \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} (2l - u - g(x)) e^{-\frac{2}{\tau}(l-g(x))(l-u)} \end{aligned}$$

Similarly to $p_\tau^x(\cdot)$, we are more interested into $q_\tau^x(\cdot)$ than into $q_\tau^x(\cdot, \cdot)$. Let's derive $q_\tau^x(u)$:

$$\begin{aligned} q_\tau^x(u) &:= \int_{g(L)}^{\min(g(x), u)} q_\tau^x(u, l) dl \\ &= -\frac{2}{\tau} \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \int_{g(L)}^{\min(g(x), u)} (2l - u - g(x)) \cdot e^{-\frac{2}{\tau}(l-g(x))(l-u)} dl \\ &= \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \end{aligned}$$

To conclude, we have the 3 equalities :

$$w(u, \tau) = e^{-\frac{1}{2}\sigma_0(u-g(x)) - \frac{1}{8}\sigma_0^2\tau}$$

$$q_\tau^x(u) = \frac{e^{-\frac{(u-g(x))^2}{2\tau}}}{\sqrt{2\pi\tau}} \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right)$$

$$C^{\text{BS}}(\tau, x) = \int_{g(K)}^{+\infty} (g^{-1}(u) - K) w(u, \tau) q_\tau^x(u) du$$

4. Computing $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(\cdot, \cdot)$

Let's look at the equation we want to prove :

$$\begin{aligned} & \mathbb{E}^{\mathbb{P}} \left[(X_T - K)_+ \cdot \mathbb{1}_{\{\min_t X_t > L\}} \right] \\ &= C^{\text{BS}}(T, x) + \int_0^T \mathbb{E}^{\mathbb{P}} \left[(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - s, X_s) \cdot \mathbb{1}_{\{\min_{0 \leq t \leq s} X_t > L\}} \right] ds \end{aligned}$$

We have to compute $(\mathcal{L} - \mathcal{L}^{\text{BS}})C^{\text{BS}}(T - s, x)$, with

$$(\mathcal{L} - \mathcal{L}^{\text{BS}})f = \frac{1}{2}\sigma_0^2(x^{2\beta} - x^2)f''(x)$$

This yields that C^{BS} has to be twice differentiable, meaning $x \mapsto w(u, \tau)q_\tau^x(u)$ must be twice differentiable. Given the form of w and q_τ^x found above, this is the case. We now have to differentiate the product of the two. We start by defining the following useful expressions :

$$U = \frac{(u - g(x))^2}{2\tau}$$

$$B = g(L) - u$$

$$A = 2g(L) - g(x) - u$$

$$C = u - g(x)$$

$$E = -\frac{2}{\tau}(g(L) - g(x))(g(L) - u)$$

4. 1. First derivatives

- We start by $\partial q_\tau^x / \partial x$:

$$\begin{aligned}
\frac{\partial q_\tau^x}{\partial x}(u) &= \frac{1}{\sqrt{2\pi\tau}} \left[\frac{\partial}{\partial x} \left(e^{-\frac{(u-g(x))^2}{2\tau}} \right) \cdot \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right. \\
&\quad \left. - e^{-\frac{(u-g(x))^2}{2\tau}} \cdot \frac{\partial}{\partial x} \left(e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right] \\
&= \frac{1}{\sqrt{2\pi\tau}} \left[g'(x) \frac{u-g(x)}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} \left(1 - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right) \right. \\
&\quad \left. - \frac{2}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} g'(x) (g(L)-u) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \right] \\
&= \frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \left(u-g(x) - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} (2g(L)-g(x)-u) \right) \\
&= \frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-U} (C - Ae^E)
\end{aligned}$$

- Then $\partial w(u, \tau) / \partial x$:

$$\frac{\partial w}{\partial x}(u, \tau) = \frac{\sigma_0 g'(x)}{2} w(u, \tau) = \frac{w(u, \tau)}{2x} \text{ because here } g(x) = \frac{1}{\sigma_0} \ln(x)$$

4. 2. Second derivatives

- $\partial_{x^2}^2 q_\tau^x$:

$$\begin{aligned}
&\triangleright \frac{\partial}{\partial x} \left(u - g(x) - e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} (2g(L) - g(x) - u) \right) \\
&= -g'(x) + g'(x) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \\
&\quad - (2g(L) - g(x) - u) \frac{2g'(x)}{\tau} \times (g(L) - u) e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \\
&= g'(x) \left[-1 + e^{-\frac{2}{\tau}(g(L)-g(x))(g(L)-u)} \left(1 - \frac{2}{\tau} (2g(L) - g(x) - u)(g(L) - u) \right) \right] \\
&= g'(x) \left[-1 + e^E \left(1 - \frac{2}{\tau} AB \right) \right] \\
&\triangleright \frac{\partial}{\partial x} \left(\frac{g'(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{g''(x)}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} + \frac{g'(x)}{\tau\sqrt{2\pi\tau}} \frac{g'(x)(u-g(x))}{\tau} e^{-\frac{(u-g(x))^2}{2\tau}} \\
&= \frac{1}{\tau\sqrt{2\pi\tau}} e^{-\frac{(u-g(x))^2}{2\tau}} \left(g''(x) + \frac{(g'(x))^2(u-g(x))}{\tau} \right) \\
&= \frac{e^{-U}}{\tau\sqrt{2\pi\tau}} \frac{g''(x) + (g'(x))^2 C}{\tau}
\end{aligned}$$

► Finally :

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} q_\tau^x(u) &= \frac{g'(x)^2 e^{-U}}{\tau\sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C \right. \\
&\quad \left. + e^E \left(1 - \frac{2}{\tau} AB - \frac{AC}{\tau} + \sigma_0 A \right) \right]
\end{aligned}$$

• $\partial_{x^2}^2 w :$

$$\begin{aligned}
\frac{\partial}{\partial x} \left(\frac{w(u, \tau)}{2x} \right) &= -\frac{1}{2x^2} w(u, \tau) + \frac{w(u, \tau)}{4x^2} \\
&= -\frac{w(u, \tau)}{4x^2}
\end{aligned}$$

4. 3. $\partial_{x^2}^2[w(u, \tau)q_\tau^x(u)]$ **computation**

We now have all the necessary elements to compute $\partial_{x^2}^2[w(u, \tau)q_\tau^x(u)]$ with Leibniz :

$$\begin{aligned}
\frac{\partial^2}{\partial x^2}[w(u, \tau)q_\tau^x(u)] &= \left(\frac{\partial^2}{\partial x^2}q_\tau^x(u) \right) w(u, \tau) + 2 \left(\frac{\partial}{\partial x}q_\tau^x(u) \right) \left(\frac{\partial}{\partial x}w(u, \tau) \right) \\
&+ q_\tau^x(u) \left(\frac{\partial^2}{\partial x^2}w(u, \tau) \right) \\
&= \frac{g'(x)^2 e^{-U}}{\tau \sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{2}{\tau} AB - \frac{AC}{\tau} + \sigma_0 A \right) \right] \\
&\times w(u, \tau) + \frac{g'(x)}{\tau \sqrt{2\pi\tau}} e^{-U} (C - Ae^E) \frac{w(u, \tau)}{x} \\
&- \frac{e^{-U}}{\sqrt{2\pi\tau}} (1 - e^E) \frac{w(u, \tau)}{4x^2} \\
&= \frac{g'(x)^2 e^{-U}}{\tau \sqrt{2\pi\tau}} \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A}{\tau} (2B + C) + \sigma_0 A \right) \right] \\
&\times w(u, \tau) + \frac{g'(x)}{\tau \sqrt{2\pi\tau}} e^{-U} (C - Ae^E) \frac{w(u, \tau)}{x} \\
&- \frac{e^{-U}}{\sqrt{2\pi\tau}} (1 - e^E) \frac{w(u, \tau)}{4x^2} \\
&= \frac{w(u, \tau) e^{-U}}{\tau \sqrt{2\pi\tau}} \left\{ g'(x)^2 \left[-1 + \frac{C^2}{\tau} - \sigma_0 C + e^E \left(1 - \frac{A}{\tau} (2B + C) + \sigma_0 A \right) \right] \right. \\
&\left. + \frac{g'(x)}{x} (C - Ae^E) - \frac{\tau}{4x^2} (1 - e^E) \right\}
\end{aligned}$$